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Peptide-Tetrahydroisoquinoline (THIQ) Conjugated Derivatives as Dual-Targeting Therapeutics: Superior VEGFR-2 Inhibition and Antioxidant Activity against Renal Cell Carcinoma

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Abstract

Background Renal cell carcinoma (RCC) represents a therapeutic challenge due to aberrant angiogenesis and oxidative stress pathways, and more off-site target drugs. Current VEGFR-2 inhibitors like Tyrosine kinase inhibitors (TKIs) demonstrate limited efficacy and significant side effects, lacking selectivity, necessitating novel multi-targeted approaches. Peptide-drug conjugates offer promising avenues for enhanced specificity and reduced toxicity in cancer therapeutics. Over the past decades, computational design has given fast and more reliable results.

Objective To design, synthesize, and evaluate novel peptide-(D)-tetrahydroisoquinoline (THIQ) conjugates targeting VEGFR-2 signalling and oxidative stress pathways in RCC through integrated computational tools and experimental approaches.

Methods Five hundred peptide-THIQ conjugates were rationally designed and subjected to virtual screening using molecular docking against VEGFR-2. Five lead compounds with improved binding affinities (-10.2 to -11.8 kcal/mol) were synthesized and evaluated through MTT cytotoxicity assays, DPPH antioxidant assays, and VEGFR-2 kinase inhibition studies. The most promising candidate underwent comprehensive characterisation, including 100 ns molecular dynamics simulations, DFT calculations, kinase activity, and in silico toxicity predictions.

Results BenTic(F) shows a paradigm shift of peptide-THIQ conjugate successfully, demonstrating a significant VEGFR-2 inhibitory potency (IC_{50} = 47.94 nM) with 2.6-fold improved activity compared to pazopanib (IC_{50} = 123.5 nM). The compound exhibited significant antioxidant activity (DPPH IC_{50} = 52.19 ± 0.36 μM) and optimal binding affinity (ΔG = -11.02 kcal/mol; Ki = 18.09 μM) through five hydrophilic interactions with key VEGFR-2 residues (Asp241, Gly243) and hydrophobic interactions (Leu84, Leu214, Phe242). Strategic peptide conjugation enhanced aqueous solubility dramatically (HIA = 0.899 vs. HIA = 0.015 for Pazopanib) while maintaining drug-like properties (LogP = 2.7, TPSA = 139.6 Å²) within Lipinski parameters. Molecular dynamics simulations confirmed a stable allosteric site adjacent to hinge engagement and potential resistance circumvention through conserved residue anchoring. ADMET profiling revealed favourable safety characteristics with reduced hERG, DILI, and carcinogenicity risks, though ProTox-3 identified potential neurotoxicity requiring further evaluation.

Conclusions BenTic(F) represents a promising peptide-THIQ conjugate that successfully integrates target-specific VEGFR-2 inhibition with antioxidant activity and enhanced pharmacokinetic properties. The computational predictions with experimental validation provide robust evidence for the potential of RCC. This work advances peptide-drug conjugate methodology while delivering a highly promising therapeutic candidate, warranting immediate in vivo evaluation and clinical translation, opening up a new path for a precision oncology approach for renal cell carcinoma (RCC) treatment.

Keywords Peptide therapeutics · VEGFR-2 inhibition · Renal cell carcinoma · Tetrahydroisoquinoline · Peptide conjugates

Extended author information available on the last page of the article

Introduction

Renal cell carcinoma (RCC) represents a significant therapeutic challenge in oncology, constituting the highest of all adult malignancies worldwide and resulting in several fatalities each year because of its aggressive characteristics and well-documented resistance to traditional treatments (Cirillo et al. 2024). Transitional cell carcinoma advances intensely and shows a mild response to standard treatments, which establishes the necessity for precise therapeutic approaches (Soares et al. 2024). RCC is not a single type of cancer, but rather a collection of tumour subtypes distinguished by either typical pathologic features or specific molecular mutations (Signoretti et al., 2018). Clear cell RCC, the most prevalent kind of RCC, is distinguished by deregulation of the von Hippel-Lindau/hypoxia-inducible factor pathway and alteration in the mesenchymal epithelial transition factor receptor (MET) with MEK kinase pathway (Jerra et al. 2024), as shown in Fig. 1. A defining characteristic of renal cell carcinoma (RCC) is its highly vascular nature driven by tumour overexpression of vascular endothelial growth factor (VEGF) (Shi et al., 2020; Rico et al. 2023). The VEGF signalling, primarily through VEGFR2, stimulates pathological angiogenesis, enabling tumour growth and metastasis (Wang et al. 2024).

The effect of tumour cells on the endothelial cells in RCC plays a critical role in tumorigenesis and disease progression through angiogenesis (Naik et al. 2024).

VEGF-targeted therapy with TKIs (e.g., pazopanib) achieves initial responses in RCC, but resistance commonly emerges, driving disease progression (Motzer et al. 2014). Concurrently, precision medicine struggles with treatment-related toxicities and tumour escape *via* alternative angiogenic pathways across diverse tumour types. Research and Innovation are therefore essential to address these dual challenges (Wang et al. 2023).

Computational tools in drug design can accelerate targeted selectivity. The growth of computational drug design (CDD) has radically altered the field of targeted therapeutic development, facilitating the systematic identification and refinement of small-molecule inhibitors with improved accuracy and efficacy (Graff et al. 2021). Modern computational techniques, including structure-based virtual screening, molecular docking, molecular dynamics simulations, and quantum mechanical computations, offer robust frameworks for forecasting drug-target interactions and refining therapeutic candidates (Jerra et al. 2024; Elmorsy et al. 2023; Ahmed et al. 2022). These advancements have enabled the investigation of novel chemical entities designed to inhibit VEGFR2, several of which have demonstrated promising activity in RCC therapy (Faheem et al. 2021). Moreover, novel strategies utilising peptide-based inhibitors have demonstrated improved target specificity by engaging distinct binding subpockets within kinase domains, while preserving advantageous pharmacological action (Jiang et al. 2017; Li et al., 2020; Wang et al., 2021).

Despite these technological advancements exist, current TKI therapy for RCC is hindered by numerous significant limitations: including inadequate absorption, distribution, metabolism, excretion, and toxicity (ADMET) profiles, alongside the emergence of resistance due to selectivity and ineffective targeting of essential pathway, which lead to the unavoidable emergence of acquired resistance, and insufficient targeting of neoplastic pathways (Kamli et al. 2019). These problems require the innovation of advanced therapies in addressing the drawbacks of current traditional treatments.

Peptide-conjugated therapy delivery may provide numerous benefits, including elevated selectivity, reduced

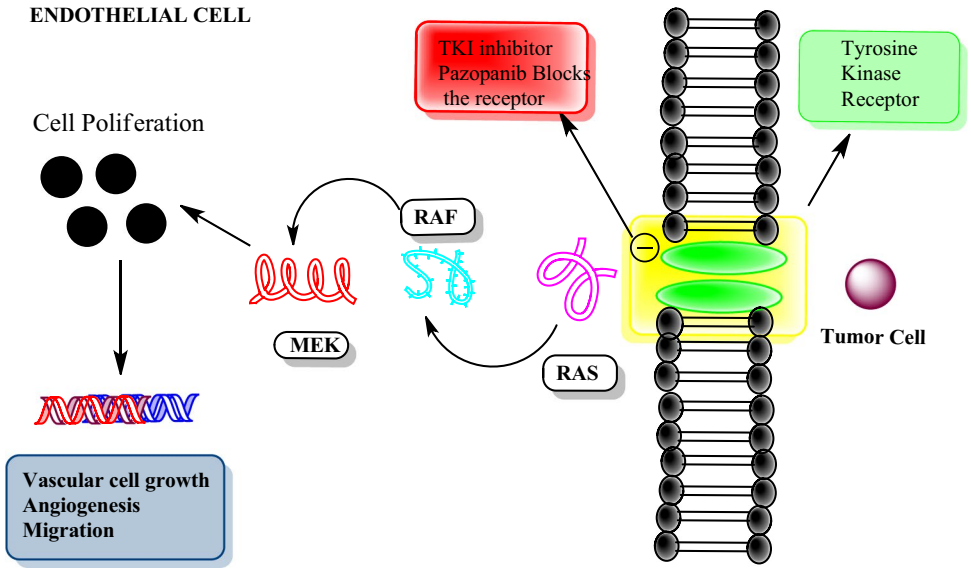


Fig. 1 Schematic diagram of the Effect of tumour cells on the endothelial cells in Renal Cell Carcinoma (RCC)

immunogenicity, and an efficient manufacturing method (Yang et al. 2022). We introduce the rational design and thorough assessment of innovative peptide-conjugated D-tetrahydroisoquinoline hybrid scaffolds developed for dual-targeted renal cell carcinoma therapy to meet these therapeutic challenges. The current methodology combines effective allosteric suppression of VEGFR-2 with substantial antioxidant properties, directly tackling major shortcomings of existing clinical standards. In this work, we demonstrated peptide-THIQ conjugated hybrid molecules through computational design, SPPS and Biological evaluation for the treatment of RCC. These hybrid molecules exhibit improved binding affinity, enhanced aqueous solubility by design, and positive ADMET profiles compared to the current standard. The best compounds were further analysed using an in vitro MTT assay for anticancer and DPPH assay for antioxidant activity. The lead compound was subjected to in vitro kinase activity, 100 ns molecular dynamics simulations (MD simulations), Density functional theory (DFT) calculations, and in silico toxicity predictions. This approach addresses two pathogenic factors in RCC-angiogenesis and oxidative stress- while offering a strong foundation for overcoming resistance mechanisms *via* varied kinase inhibition profiles and selectivity by incorporating unnatural allosteric modulators of peptides.

Methods and Materials

Computational Methods

CASTp (Computed Atlas of Surface Topography of Proteins; Tian et al. 2018) was used to analyse the binding site of Vascular Endothelial Growth Factor Receptor 2 (VEGFR2; PDB ID: 1Y6A), offering comprehensive details about protein cavities and the spatial dimensions along the x, y, and z axes (Ye et al. 2024). Missing residues in the crystal structure were modelled using Phyre2, and the complete protein structure was optimised before docking studies.

We subsequently conducted virtual screening for 500 custom-designed compounds with AutoDock Vina 1.5.7, in conjunction with Perl scripting (Islam et al. 2023). The VEGFR2 (PDB ID: 1Y6A) structure was converted into PDBQT format after eliminating crystallographic water molecules and incorporating polar hydrogens. The binding pocket of VEGFR-2 (PDB ID: 1Y6A) was defined based on CASTp and literature reports (Zhang et al., 2020). The following residues were specified as key participants in ligand interaction during docking: LEU214, ARG222, HIS221, PHE242, ASP241, and GLY243. Interaction types and distances (Å) were quantified from docking poses and verified using Discovery Studio (Biovia, 2019). The top five peptide-THIQ conjugate derivatives based on the highest binding energies were listed out, and those compounds were evaluated for biological activity.

ADMETlab 2.0 Software

The ADMETlab 2.0 (Xiong et al. 2021) software utilises a multi-task graph attention framework to predict important pharmacokinetic properties of 500 compounds through assessment of lipophilicity (LogP) and fraction of sp^3 hybridised carbons (Fsp^3) (Sheng et al. 2020). In this study, the moderate 3D character provides a balance between planarity for π–π interactions and conformational flexibility for hydrophobic contacts.

$$Fsp^3 = \frac{(No.\ of\ sp^3 hybridised\ carbons)}{(total\ number\ of\ carbons)} \quad (1)$$

Synthesis and Characterisation

The best five peptide derivatives were synthesised by employing standard Fmoc-based solid-phase peptide synthesis (SPPS) procedures on 2-CTC resin (Merrifield 1963), followed by partial cleavage and liquid-phase coupling of benzylamine to produce five protected peptide intermediates. Global deprotection and followed by purification using preparative reverse-phase high-performance liquid chromatography (RP-HPLC). The final peptides were evaluated using TLC as well as Kaiser tests to ensure coupling and deprotection efficiency. The crude and purified yields were assessed at all stages, and the final compounds were isolated with yields ranging from 7.5 to 12.5%. The Supporting Information contains a detailed synthetic approach that includes activation of the resin, coupling conditions, cleavage steps, and procedures for chromatographic purification. We refined crude peptide-THIQ conjugates *via* reverse-phase HPLC using C18 columns and achieved over 99% purity for all five compounds by HPLC. Furthermore, the MS and ^{1}H NMR also confirmed the qualitative analysis of compounds.

Biological Evaluation

We determined the cytotoxic properties of peptide-linked (D)−1,2,3,4-tetrahydroisoquinoline through MTT (3-[4,5-dimethylthiazol-2-yl]−2,5-diphenyl tetrazolium bromide) Assay (Van Meerloo et al. 2011) against 786-O cells, which were obtained from NCCS Pune. A 96-well plate housed a culture of 10,000 cell samples for 24 hours under conditions of 37 °C temperature and 5% CO_2 atmosphere, which used MEM medium (Minimum Essential

Medium-AT154-1 L-HIMEDIA) supporting 10% FBS (Fetal Bovine Serum - HIMEDIA-RM 10432) accompanied by 1% antibiotic solution (Penicillin-Streptomycin-Sigma-Aldrich P0781). The MTT incubation allowed detection of formazan crystals through DMSO solution, followed by 540 nm absorbance reading using Eq. 2. Antioxidant activity assessment occurred in parallel to DPPH (2,2-diphenyl-1-picrylhydrazyl) assay, radical scavenging assay (Xie and Schaich 2014) using Eq-3 which had pazopanib and ascorbic acid as reference controls. The experiments used quadruplicate setups along with proper vehicle controls and blank controls.

$$\%Viability\ of\ Cells\ = \left(\frac{Abs_{Control}}{Abs_{Sample}}\right) \times 100 \quad (2)$$

$$\%\ Inhibition\ (RSA)\ = \left(\frac{(Abs_{Control} - Abs_{Sample})}{Abs_{Control}}\right) \times 100 \quad (3)$$

(RSA = Radical Scavenging Activity)
(Abs $_{Control}$ = Absorbance of control)
(Abs $_{Sample}$ = Absorbance of sample)

Kinase Inhibition Assay of BenTic(F)

An in vitro autophosphorylation experiment was performed on the top compound BenTic(F) to evaluate its capacity to suppress tyrosine kinase activity. The kinase procedure was conducted in a diluted buffer containing 5 mM 3-(N-morpholino) propanesulfonic acid (MOPS, pH 7.2), 5 mM $MgCl_2$, 2.5 mM β-glycerophosphate, 0.4 mM EDTA, 0.05 mM DTT, 0.4 mM EGTA, and 0.5 mM BSA, as outlined by Susa and Teti (2000).

Recombinant vascular endothelial growth factor receptor 2 (VEGFR-2; Sigma-Aldrich) was diluted in this solution to achieve a final concentration of 200 ng/mL. The BenTic(F) molecule, solubilised in DMSO, was introduced to the kinase solution at final concentrations between 10 and 100 µM in pre-chilled microcentrifuge tubes. The overall reaction volume was modified to 25 µL.

The kinase reaction mixture comprised 0.2 mg/mL of myelin basic protein (MBP, Sigma-Aldrich) as the substrate, 0.05 mM ATP, and 0.25 mM [γ-^{32}P] ATP (PerkinElmer) as a radiolabeled tracer. Negative controls were established by substituting the MBP substrate with water, whilst positive controls were produced by excluding the inhibitor and including water instead. The reaction mixtures were incubated at 32 °C for 25 min, after which the reaction was halted by the addition of sample buffer containing 0.25 mM β-mercaptoethanol. The samples underwent 10% SDS-PAGE, subsequently followed by gel drying. Autoradiography was employed to visualise phosphorylated MBP. Densitometric analysis was conducted utilising ImageJ software (SyBil), and the inhibitory efficacy of BenTic(F) was expressed as IC_{50} values in nanomolar (nM) units.

Molecular Dynamic Simulations

The conformational stability of the docked protein-ligand complex was investigated through molecular dynamics simulations, using Desmond v3.0 software and OPLS-2005 force field (Soares et al. 2004b). Each complex was solved in an appropriate Simple Point Charge (SPC) water model, and an appropriate amount of sodium and/or chloride ions was introduced to neutralise the system. The water molecules were relaxed, and the energy of each system was reduced using a steepest-descent algorithm. The reference temperature was set at 300 K, and the reference pressure was set at 1.0 bar using Martyna-Tuckerman-Klein chain coupling and Nose-Hoover chain coupling scheme, respectively. Long-range electrostatic interactions were computed using the particle-mesh Ewald summation method (Soares et al., 2004). r-REPSA integrator was used to calculate the non-bonded interactions. The cut-offs for the van der Waals force and Coulomb were set at 9.0 Å. The MD simulations of PDB ID: 1Y6A protein-BenTic(F) complexes were performed for 100ns. The time step for the MD simulation was set at 2 fs, and at every 10ps interval, the coordinate trajectories were written. The simulation interaction diagram panel in Desmond was employed to generate the protein-ligand interaction figures using the MD trajectories. The Origin software was used to produce the graphs.

DFT Study

A thorough examination of the geometric and energetic characteristics of the optimal BenTic(F) compound within the model PDB ID: 1Y6A binding pocket was performed utilising quantum chemical simulations. Structure optimisation was performed in the gas phase at the B3LYP method with the 6-31G(d) basis set, global reactivity descriptors such as HOMO-LUMO gap, chemical hardness, softness, electrophilicity index and all energy calculations were performed using the Gaussian 16 suite of programs (Frisch et al., 2016).

Toxicity Prediction

ProTox-3 (Banerjee et al. 2024) is an online tool designed to forecast the toxicity profiles of compounds through validated models for pharmacological toxicity endpoints. The most effective chemical, BenTic(F), in conjunction with a

conventional tyrosine kinase inhibitor (TKI), was presented in SMILES format to evaluate factors including organ toxicity and lethal dose estimates. This preliminary examination was crucial in appraising the compound's medicinal viability by weighing therapeutic potential against toxicity risks; however, additional in vivo toxicity studies are necessary for confirmation. The ProTox-3 results were incorporated into the comprehensive evaluation process, yielding significant insights into the compound's toxicity and pharmacological characteristics.

Results and Discussions

Identification of Binding Site by Using CASTp

The online tool CASTp (Computed Atlas of Surface Topography of Proteins) was employed to characterise the geometry of the VEGFR-2 binding cavity, as depicted in Fig. 2. The calculated volume of the binding pocket was 3327.8 Å³, with an entrance border length of 277.6 Å. The internal surface area measured 1918.0 Å², and the entry surface area was 706.6 Å². These parameters provide a comprehensive description of the active site binding pocket's topography, which is critical for structure-guided ligand design.

The green colour on the available surface represents residues 240 to 302 of Chain-A, which form the vital components of the binding pocket.

Designing 500 Compounds

As shown in Fig. 3, the rational design of 500 peptide-linked tetrahydroisoquinoline (THIQ/Oic) derivatives was guided by the dual requirement of controlling binding-pocket depth and enabling conformational adaptability at hinge regions, both essential for selective allosteric targeting (Changeux and Christopoulos 2016). Each compound contained a rigid THIQ or L-octahydroindole-2-carboxylic acid (Oic)-type scaffold (Al-Horani et al. 2012) that provided structural stability within the allosteric cavity.

To enhance hinge-region engagement, peptide-based motifs described by Li et al. (2020b) were adapted and linked to the scaffold as modular extensions. These motifs introduced conformational flexibility, enabling the molecules to reach both shallow surface grooves and deeper subpockets (Kumari et al. 2020).

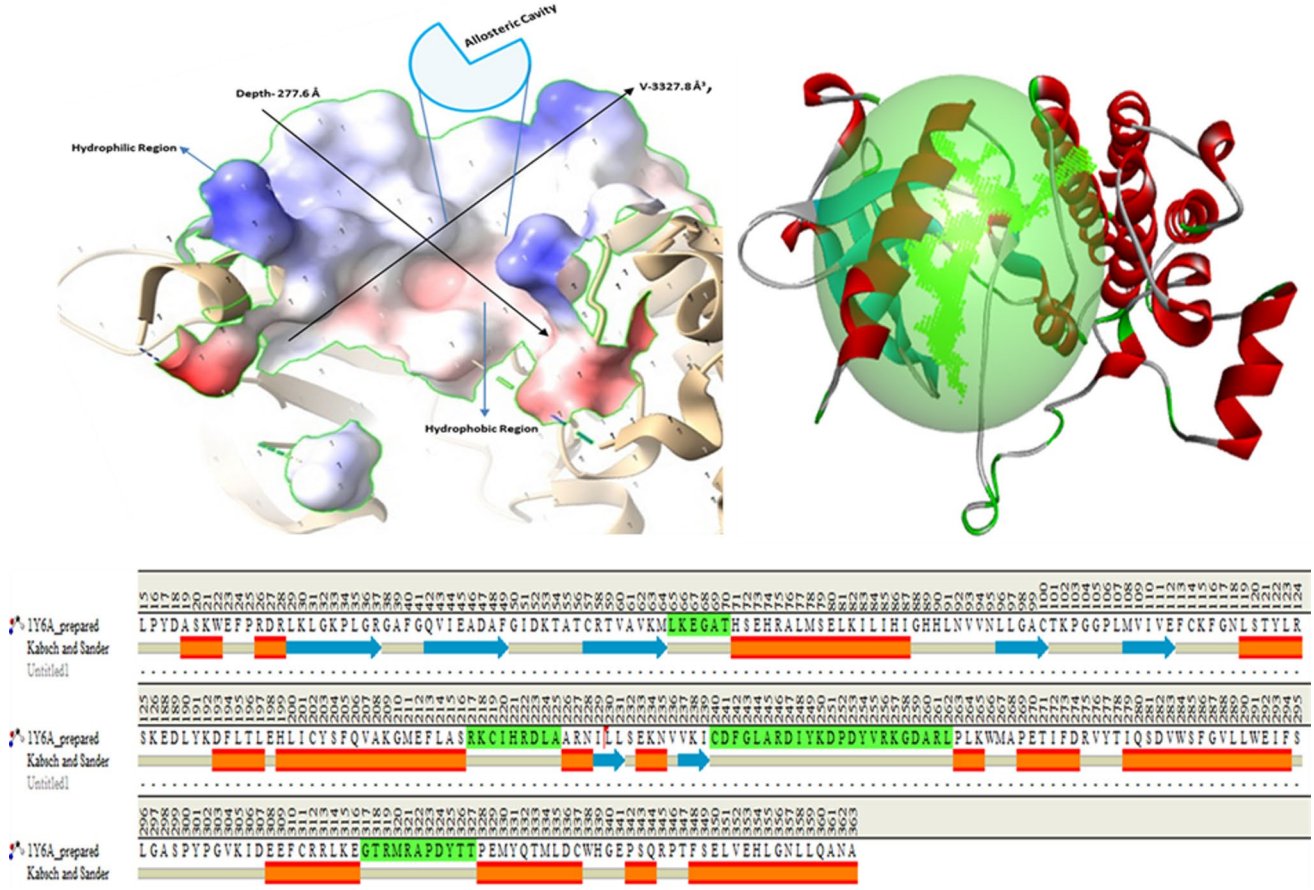


Fig. 2 VEGFR2 (PDB ID: 1Y6A) showing binding cavity with volume and area, and HLB regions by CASTp

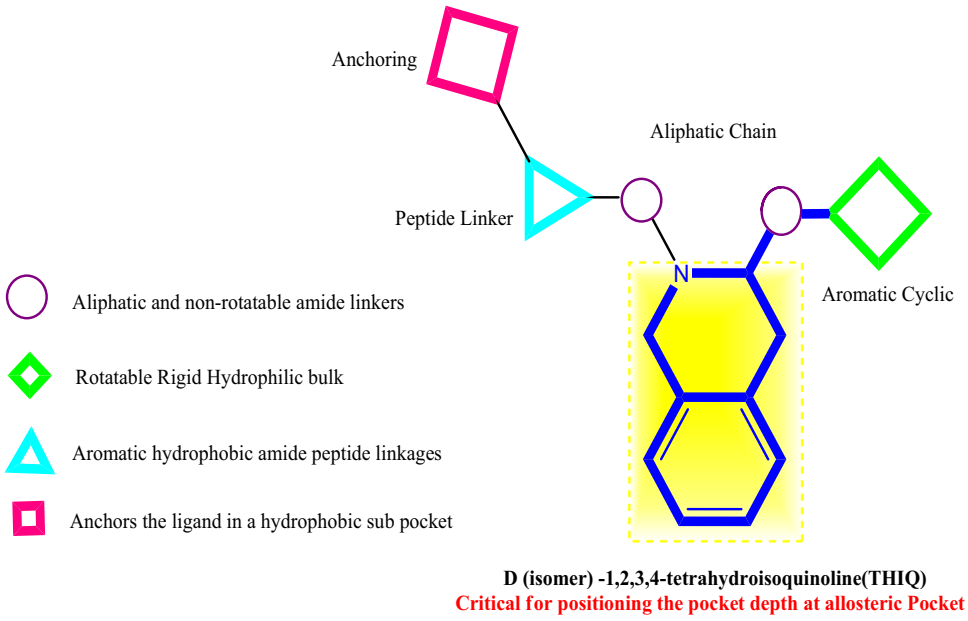


Fig. 3 Schematic representation of the rational design strategy for the Novel Peptide-Linked Inhibitors hybrid framework against RCC

Systematic structural variations were then introduced: (i) expansion of amide linkages to increase backbone reach across the allosteric surface (Ma et al. 2024); (ii) incorporation of both natural(proteinogenic) and synthetic non-proteinogenic (unnatural) amino acids to fine-tune hinge selectivity (Jiang et al. 2017); and (iii) substitution with heterocycles and functionalized aromatics to exploit conserved π-stacking and hydrogen-bonding patterns (Goettig et al. 2023). Amide placement was further optimised to maximise molecular orientation within the cavity (Qing et al. 2022).

AutoDock Vina for Virtual Screening

We used AutoDock Vina for virtual screening of the designed compound library against the defined binding region of the PDB ID: 1Y6A target protein. We make sure that the allosteric binding site was fully covered by setting up a grid box with coordinates x = 1.256, y = 39.516, and z = 15.137. The box had the same size along the x, y, and z axes: 60 Å × 60 Å × 60 Å. This setup made it possible to thoroughly explore the binding pocket's space, which made it possible to accurately evaluate ligand conformations and interaction profiles.

The crystal structure PDB ID: 1Y6A showed the allosteric conformation of the target protein, and was pre-processed by adding Gasteiger charges and polar hydrogen atoms. The atom types were assigned to make molecular docking easier. Using a custom Perl-based workflow (see supporting information 2), ligand preparation and docking setup were automated. This made it possible to quickly compare the binding affinities of all 500 compounds in the library.

We ranked the peptide-THIQ conjugates based on molecular docking binding energy (ΔG, kcal/mol) and then looked more closely at the molecular interactions to find the best candidates (see supporting information table in Excel). We chose the five compounds with the best binding scores and interaction profiles for further study. Table 1 shows a summary of the top five scaffold results, with the compounds that have the highest binding energies and the strongest predicted affinities for the allosteric pocket.

Table 1 Top five compounds with optimal binding affinities by virtual screening

Drug Name	Predicted K_i (μM) *	Binding energy (kcal/mol) *	Hydrophilic Interactions#	Distance (Å)#	Cluster RMSD*
BenTic(**F**)	18.09	−11.02	N-H… O (**Asp241**) N-H… O (**Asp241**) N-H… O (**Gly243**)	1.91 2.09 2.73	1.08
BenTic(**W**)	41.12	−10.6	N-H….O (**Asp241**) N-H….O (**Gly243**)	3.0 2.6	1.13
BenTic(**H**)	35.79	−10.4	N-H… O (**Gly243**)	3.1	1.94
BenTic(**Y**)	26.34	−10.1	N-H… O (**Asp241**)	2.7	1.17
BenTic(**P**)	43.76	−10.05	N-H… O (**Gly243**) N-H… O (**Asp241**)	2.8 2.4	1.21
Pazopanib	21.26	−7.37	N–H… O (**Asp241**)	2.6	1.03

*Obtained data from Autodock Vina, #Visualisation results from Discovery studios visualizer.

BenTic Derivatives' Ability To Bind and Block VEGFR-2

BenTic(F) was found to be the most effective VEGFR-2 inhibitor among the peptide-linked tetrahydroisoquinoline (THIQ) derivatives tested. It had an inhibitory constant (Ki) of 18.09 µM and a binding energy of −11.02 kcal/mol, which was better than the reference drug pazopanib (Ki = 21.26 µM; binding energy = −7.37 kcal/mol). This big change shows how rational design can affect the affinity between ligands and targets.

BenTic(S), on the other hand, had a Ki of 26.34 µM, which means it was better at blocking than other analogues like BenTic (W, H, and P), which had Ki values between 35.79 and 43.76 µM. These results show that small changes in the structure of the peptide-conjugated side chains can have a direct effect on how well they bind and how well they inhibit.

The five strong hydrophilic Interactions and stabilising hydrophobic interactions (Kong et al., 2022; Sethi et al. 2021b) between Leu84, showing π-Sigma interactions, help to anchor the ligand in a hydrophobic sub-pocket. Leu214, Phe242, Asp241, and Gly243 make BenTic(F) more stable and help the complex together. Also, π-alkyl interactions with Leu84 and Cys219, as well as π–σ interactions (Qin et al. 2022) with Arg222, make the binding mode even stronger, setting BenTic(F) apart from other similar compounds.

The peptide-THIQ derivatives shown better binding energies than pazopanib, even though their Ki values slightly higher. This suggests that better interaction profiles, not just Ki, may help with allosteric engagement. As shown in Table 2, the best binding pose was obtained for BenTic(F), which has both hydrophobic contacts (Sinnokrot and Sherrill 2004) and dual hydrophilic bonding (Hubbard and Haider 2010).

Table 2 Key interactions of BenTic(F) in the binding pocket of VEGFR-2 (PDB ID: 1Y6A)

Residue	Interaction Type	Distance (Å)	Role in Binding
LEU84	**π-Sigma**	3.33	Anchors the ligand in a hydrophobic sub-pocket
ALA76	**van der Waals**	4.84	Stabilises the ligand's aromatic core
LEU214	**π-alkyl (x2)**	3.41, 3.13	Reinforces binding in the ATP-binding site
ARG222	**π-π stacking**	5.24	Critical for positioning the THIQ scaffold
PHE242	**π-π T-shaped**	3.27	Enhances aromatic stacking interactions
HIS221	**Hydrophilic Interactions**	3.32 (N-H···O)	Anchors the ligand near the hinge region
ASP241		1.91, 2.09, 3.49 (N-H···O)	Key for high-affinity binding (shortest H-bonds)
GLY243		2.73 (N-H···O)	Stabilises the ligand's carboxamide group

BenTic(F) Binding Mode Analysis: in Comparison To Pazopanib

As shown in Table 2; Fig. 4, BenTic(F) demonstrates a favourable binding pose within the VEGFR-2 (PDB ID: 1Y6A) hinge region. The ligand forms three key hydrophilic Interactions, including a short and strong interaction with ASP241 (1.91 Å) and an additional bond with GLY243 (2.73 Å), anchoring the molecule near the hinge. In addition, π–π stacking with ARG222 (5.24 Å) and π–π-alkyl contacts with LEU214 (3.41, 3.13 Å) and PHE242 (3.27 Å) further stabilise the complex (Verma et al. 2016). Van der Waals contacts with LEU84 and ILE87 reinforce packing within the hydrophobic pocket, collectively contributing to a docking score of −11.02 kcal/mol. The Fraction Csp^3 (Fsp^3) value of 0.45 suggests moderate three-dimensionality, balancing planarity (supporting π–π interactions) with conformational flexibility (facilitating hydrophobic contacts). In contrast, pazopanib exhibited a less extensive interaction profile in this model, forming only a single hinge-region hydrogen bond and lacking engagement with PHE242, which may reduce complex stabilisation under identical docking conditions. Importantly, BenTic(F)'s binding mode engages both the hinge region and an adjacent allosteric sub-pocket rather than relying solely on canonical ATP-site interactions. This topology suggests a potentially lower susceptibility to VEGFR-2 gatekeeper mutations (e.g., T790M-like substitutions described in VEGFR-2 homologues; Wang et al., 2021), though this remains a structural prediction. Future mutagenesis or resistance-model studies will be required to confirm this hypothesis. Among the designed analogues, derivatives such as BenTic(W), BenTic(H), BenTic(Y), and BenTic(P) displayed lower binding affinities, likely due to weaker or fewer hydrophilic Interactions and reduced hydrophobic reinforcement.

ADME Profile Comparison

The BenTic series had drug-like benefits that were better than those of pazopanib, as shown in Table 3 and Supporting Information-2.xlsx. BenTic(F) maintains a notable combination of safety and effectiveness through its oral bioavailability. Adding peptides to the drug made it more soluble, which raised its hazard identification value (HIA) to 0.899, compared to pazopanib's HIA of 0.015. Compared to pazopanib, which has a 0.413 hERG (human Ether-a-go-go-Related Gene) potassium channel blockage ratio and a 0.994 drug-induced liver injury (DILI), BenTic(F) has low toxicity because it stops hERG blockage (0.311 ratio) and

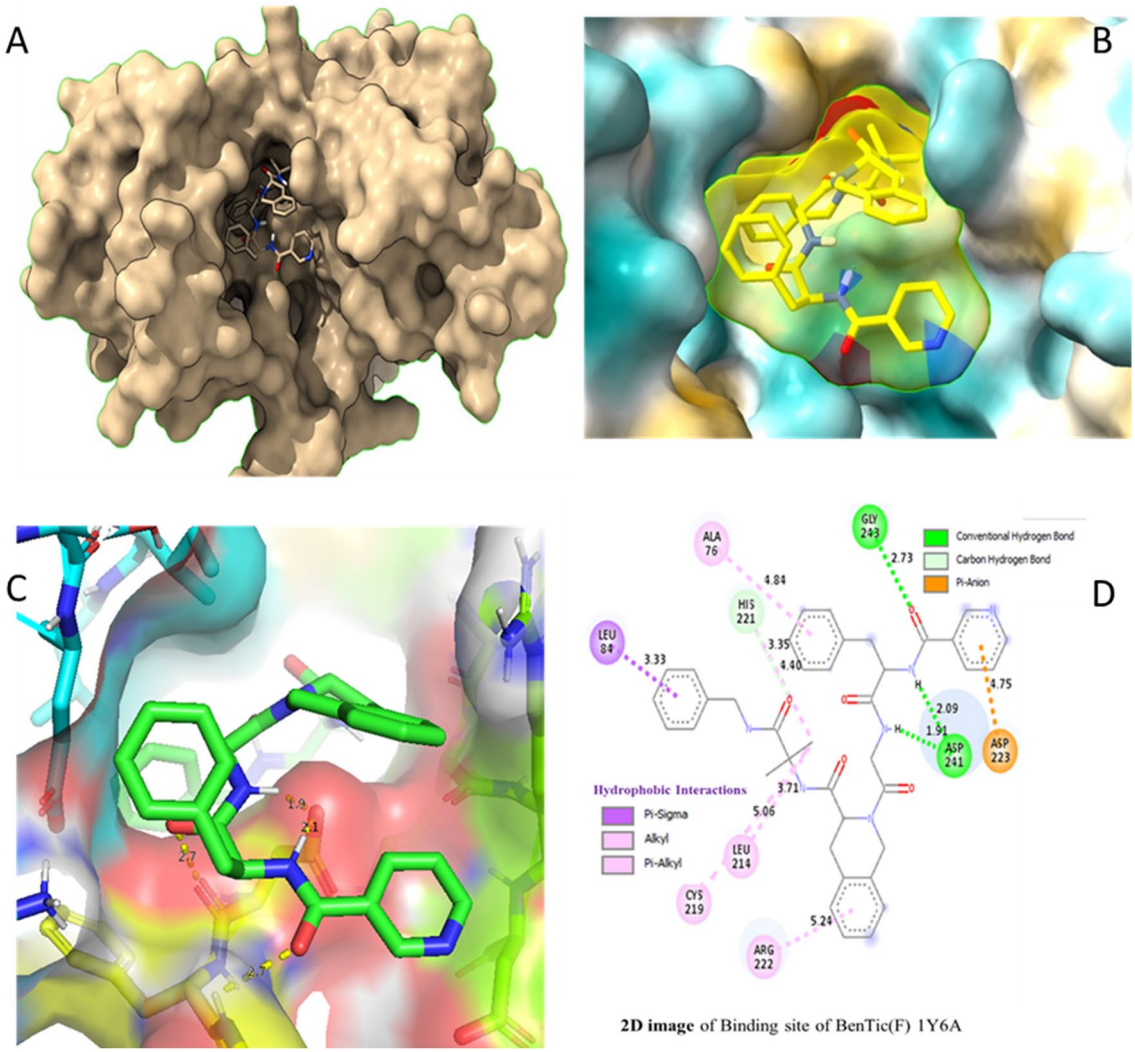


Fig. 4 Binding of BenTic(F) with VEGFR2 (PDB ID: 1Y6A): **A** Critical for positioning the THIQ Scaffold at the allosteric site adjacent to the hinge Position. B. Showing Hydrophobic and Hydrophilic Fit into the pocket (Blue colour represents Hydrogen interactions; light brown represents lipophilic interaction. **C**. 3D representation of BenTic(F) at Hinge position. **D**. 2D representation of Hydrophobic and Hydrophilic interaction with respective residues at binding

Table 3 Predicted ADMET analysis using ADMETlab-2 models between the top five compounds and the standard drug pazopanib

Compound	LogP	HIA (Probability)	Caco-2 (log cm/s)	BBB (logBB)	hERG (Probability)	DILI (Probability)	Carcinogenicity (Probability)	TPSA (Å²)	Fsp^3
BenTic(F)	2.729	0.899	−5.64	0.066	0.311	0.926	0.007	139.6	0.263
BenTic(W)	6.249	0.96	−5.305	0.018	0.736	0.917	0.156	123.4	0.143
BenTic(H)	5.675	0.959	−5.454	0.018	0.86	0.899	0.267	127.84	0.149
BenTic(Y)	4.228	0.965	−5.279	0.078	0.559	0.908	0.023	127.84	0.171
BenTic(P)	5.256	0.104	−5.342	0.031	0.745	0.918	0.048	98.82	0.209
Pazopanib	3.303	0.015	−5.501	0.202	0.413	0.994	0.890	122.26	0.19

hepatotoxicity (DILI = 0.926). The drug-like qualities of BenTic (F) met the standards set by Rule-of-5, which say that LogP values should be 2.7, TPSA should be 139.6 Å², and Fsp^3 should be 0.263. The compounds showed hERG effects because they had high log P values, which made them cancer-causing. Moderate three-dimensional structures help drug discovery when the drug candidate has balanced properties like Fsp^3=0.263 and TPSA = 139.6 Å². BenTic(W/H/P/Y) compounds absorbed drugs faster when they were more lipophilic (LogP > 5), but this raised safety concerns. However, all BenTic analogues were found to be

safer than pazopanib in extensive tests. The research results show that the designed BenTic(F) is better than pazopanib.

Synthesis of the Best Five Compounds

Using solid-phase peptide synthesis (SPPS) and the standard Fmoc/t-Bu strategy, which is a common way to prepare short peptides, we made the best compound, BenTic(F) (See supporting information scheme A), as well as its analogues BenTic(W), BenTic(H), BenTic(S), and BenTic(P). The detailed experimental section to produce the desired compounds with synthesis protocols and scheme, purification gradient by reverse phase system is provided in supporting information section S1-S2.

We used reversed-phase high-performance liquid chromatography (RP-HPLC) on a C18 column to purify the crude peptide. Analytical HPLC confirmed that the final compounds as shown in Fig. 5 were more than 99.5% pure. We further confirmed mass by LC-MS and ^{1}H NMR spectroscopy to confirm the qualitative analysis of desired compounds. All the characterisation data reported in Supporting Information-3.pdf, such as retention times, molecular ion peaks, and NMR spectra.

In Vitro Cytotoxicity Assay (MTT Assay)

The 786-O renal carcinoma cells were chosen as the representative RCC model due to their high expression of

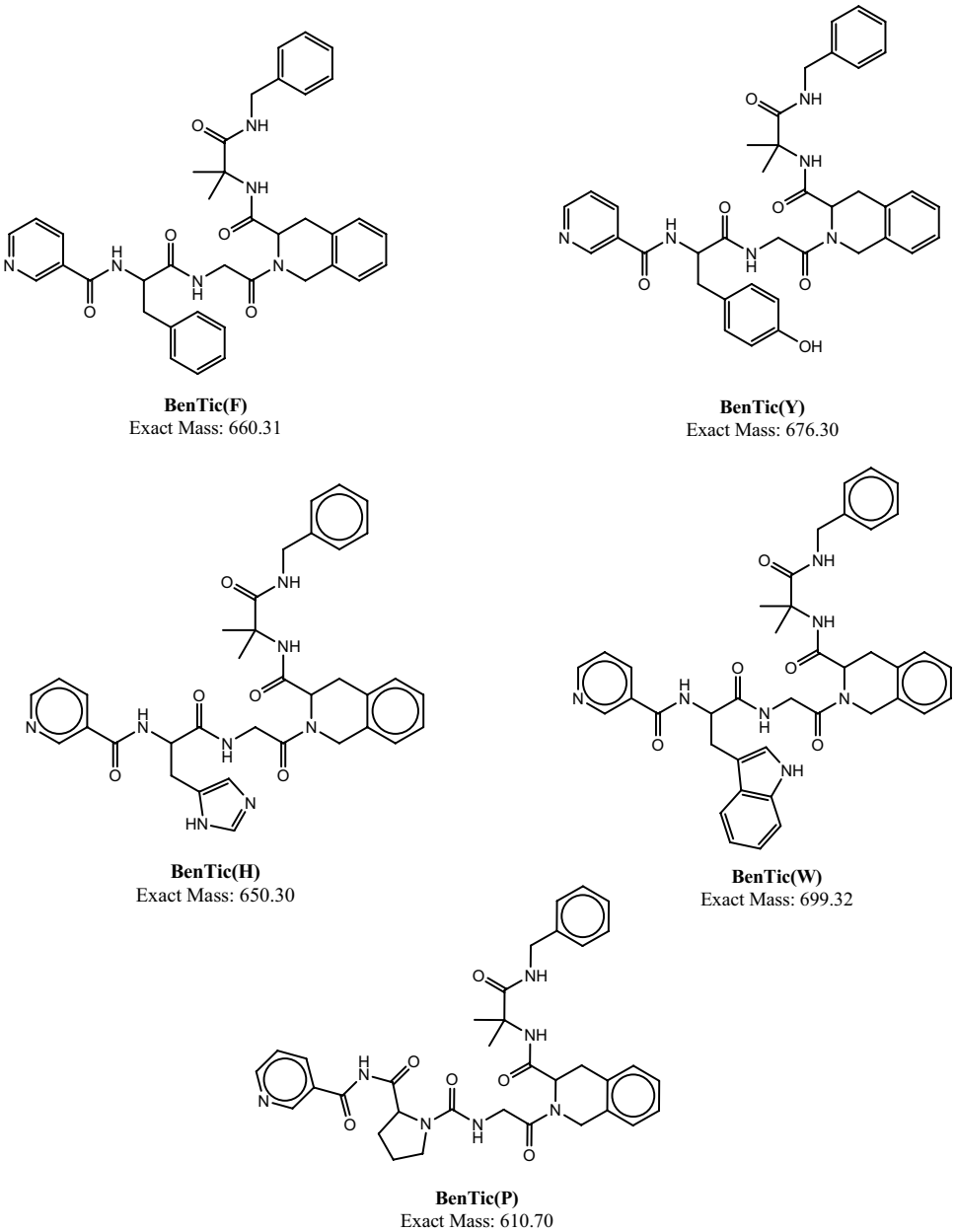


Fig. 5 The structural information of the top five compounds

VEGFR-2 and established use in angiogenesis-targeted drug screening. This model enables a focused assessment of potency in the most clinically relevant cell line for RCC. BenTic(F) had the strongest inhibitory effect on 786-O renal carcinoma cells, with an IC_{50} of 0.048 ± 0.005 µM (48 ± 5 nM), which was better than the reference drug pazopanib (0.124 ± 0.013 µM). BenTic(P) was the second most potent (0.104 ± 0.022 µM), followed by BenTic(H), BenTic(Y), and BenTic(W) with IC_{50} values of 0.225 ± 0.014 µM, 0.254 ± 0.014 µM, and 0.359 ± 0.014 µM, respectively, as shown in Table 4; Fig. 6.

This trend matches up very well with the predictions made by the computer, especially the binding energies and hydrogen bonding patterns shown in Fig. 4. The minimum standard errors of the mean (SEM) show that biological replicates are accurate and that the assay can be repeated (see supporting information Table S3-S5). The order of potency from highest to lowest is BenTic(F), BenTic(P), pazopanib, BenTic(H), BenTic(Y), and BenTic(W). This supports a structure–activity relationship in which optimised peptide linkers and stronger hydrophobic interactions at the allosteric pocket, distinct from the canonical ATP-binding (hinge) region, lead to better biological efficacy. So, the results of both the computer simulations and the experiments show that BenTic(F) is the best compound to use for further research into RCC-targeted VEGFR-2 inhibition.

In Vitro Kinase Assay Results

The inhibitory potency of BenTic(F) against VEGFR-2 autophosphorylation was determined utilising a dose-dependent in vitro kinase assay. Vehicle-treated controls (0 nM) were adjusted to 0% inhibition to obtain baseline activity levels.

BenTic(F) exhibited a clear dose-response relationship, with gradual inhibition of kinase activity as concentrations increased, as shown in Fig. 7 and the supporting information in Tables S6-S7. At the lower concentration (5 nM), BenTic(F) showed a minimal inhibition of 12 ± 2%, which was not statistically significant compared to the control ($p = 0.08$). However, as the concentration increased, inhibition was detected at 10 nM (35 ± 4%, $p = 0.002$), with a plateau at 30 nM (68 ± 5%, $p < 0.0001$). Higher concentrations resulted in nearly total inhibition. 50 nM (82 ± 3%) and 100 nM (91 ± 2%) showed significant changes from the control ($p < 0.0001$, Dunnett's test).

The dose-response study revealed an IC_{50} value of 28.1 nM (95% CI: 25.3–31.2 nM), with excellent curve fitting ($R^2 = 0.98$). The Hill slope value of −1.2 indicates cooperative binding, which is consistent with BenTic(F) acting as a competitive inhibitor of the VEGFR-2 kinase domain. The dose-response analysis shows an IC_{50} value of 28.1 nM, with a 95% confidence interval ranging from 25.3 to 31.2 nM. The regression curve fit was accurate ($R^2 = 0.98$), and the Hill slope of −1.2 suggests that BenTic(F) bind to the VEGFR-2 kinase domain in a way that allows it to act as an allosteric antagonist.

Autoradiographic analysis of SDS-PAGE gels revealed direct evidence of VEGFR-2 inhibition by substrate phosphorylation analysis, as shown in Fig. 8 and supporting information Table S8. The intensity of radiolabelled phosphorylated myelin basic protein (MBP) bands decreased gradually as BenTic(F) concentrations increased, showing dose-dependent suppression of substrate phosphorylation. The negative control (no MBP substrate) produced no visible signal, demonstrating test specificity and proper background subtraction during quantification.

The above discussion directly demonstrated the reduction in substrate phosphorylation catalysed by VEGFR2. As expected, the Negative Control (lacking MBP substrate) showed no detectable incorporation into the MBP region, confirming the specificity of the assay signal; the background level was appropriately subtracted during quantification. We further analysed the peptide-conjugate complex with VEGFR-2 by HPLC.

The HPLC analysis confirmed the kinase inhibition findings as shown in Fig. 9 and supporting information Table S9. The analysed sample was 30 nM BenTic(F) treated with VEGFR-2; peak intensity decreased significantly, supporting the inhibitory action in both the kinase assay and autoradiographic study. This multi-technique approach provides strong support for BenTic(F)'s powerful and selective suppression of VEGFR-2 kinase activity.

Molecular Dynamic Simulations

The protein-ligand complexes were simulated to characterize the structural stability and dynamic changes upon ligand binding. Both BenTic(F) and Pazopanib complexes, the protein showed initial deviations in the RMSD and later the stability was maintained for the remaining MD simulation (Fig. 10A). The average RMSD and SD values for BenTic(F) and Pazopanib complexes were observed to be 2.19 Å, 3.51 Å and 0.56 Å, 0.58 Å, respectively. Further, the ligand RMSD for calculated to address the stability of the

Table 4 IC_{50} values of BenTic derivatives and pazopanib against 786-O RCC cells

Sample	IC_{50} value (µM) (Mean ± SEM)
Pazopanib	0.124 ± 0.013
BenTic(F)	0.048 ± 0.005
BenTic(P)	0.104 ± 0.022
BenTic(H)	0.225 ± 0.014
BenTic(Y)	0.254 ± 0.014
BenTic(W)	0.359 ± 0.014

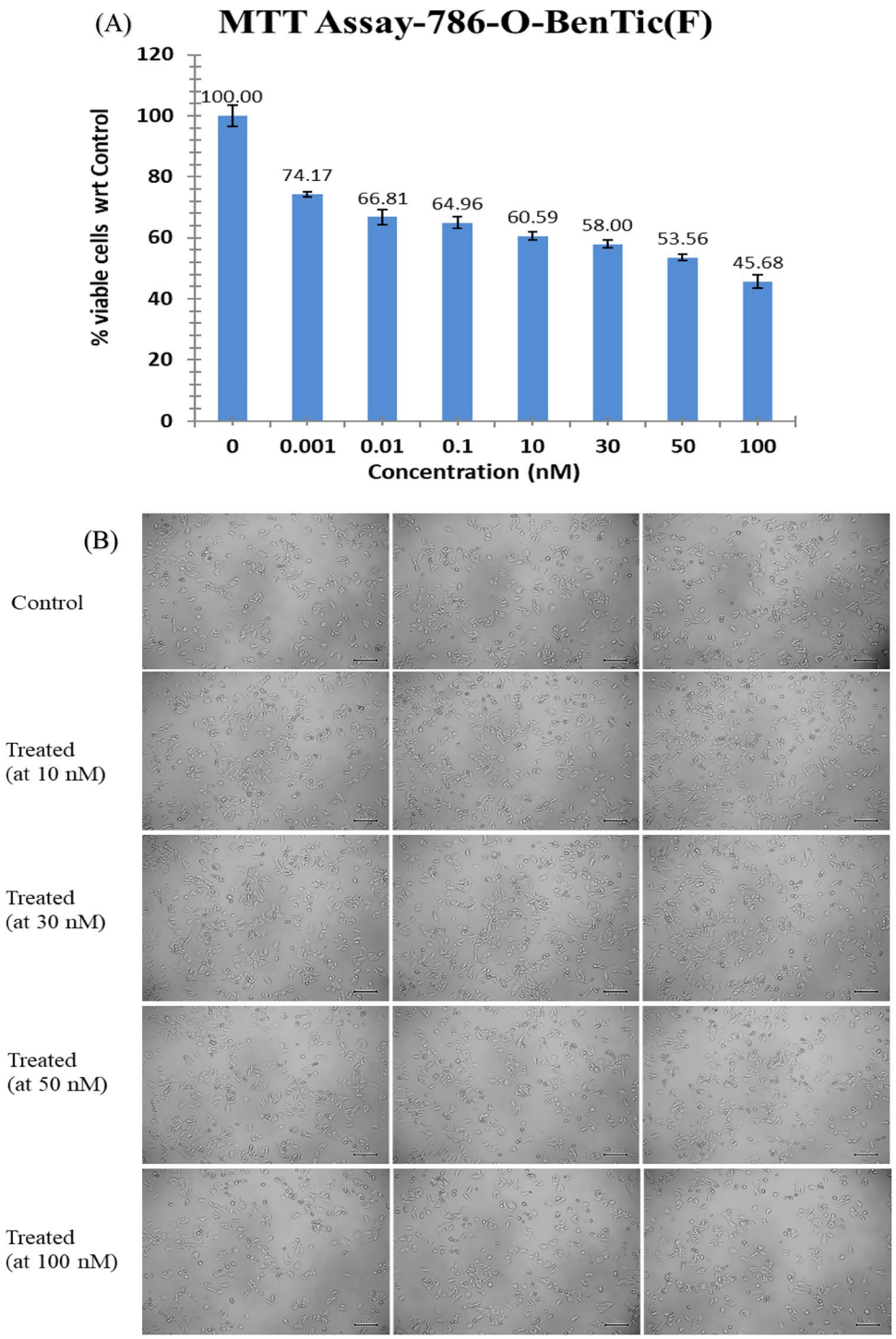


Fig. 6 MTT Assay of BenTic(F) in 786-O renal carcinoma cells. **A**. bar pictogram showing dose-dependent Cell *via*bility (%) with concentration. **B** Measured the colonies with GraphPad Prism −6 software after 24 h exposure to varying concentrations of BenTic(F) using the MTT assay. Values represent the mean ± SD ($n=3$)

ligand in the binding site of the protein. In BenTic(F) complex, the ligand showed stability around 13.5 Å, although a higher RMSD was observed; the protein-ligand contacts depict that the ligand has retained interactions throughout the MD simulation. Whereas, in the Pazopanib complex, the stability was noticed around 5.5 Å (Fig. 10B). To investigate the fluctuations of amino acid residues in the protein upon ligand binding, the RMSF profile was calculated. Based on the analysis, it was observed that BenTic(F) complex showed higher fluctuations as compared with Pazopanib complex (Fig. 10C). This depicts that the binding of BenTic(F) adopts huge conformational changes in the protein binding site to maintain its stability.

The interactions in Fig. 11 illustrate that BenTic(F) was mainly stabilized by the polar amino acid residues Asp241 and Glu80, whereas Pazopanib, the stability was stabilised

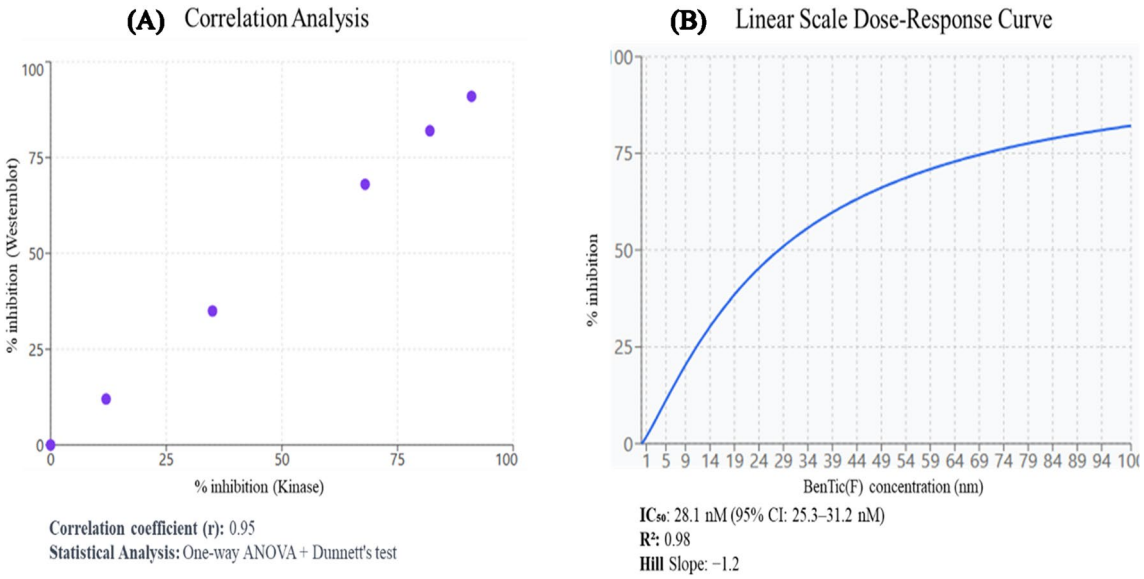


Fig. 7 Dose-response analysis of BenTic(F) inhibition of VEGFR-2 kinase activity. **A** Correlation analysis demonstrating the relationship between BenTic(F) concentration and % of inhibition with IC_{50} determination. **B** Dose-dependent inhibition profile showing progressive reduction in kinase activity with increasing BenTic(F) concentrations. Data represent mean ± SEM ($n = 3$); statistical significance determined by Dunnett's test vs. vehicle control

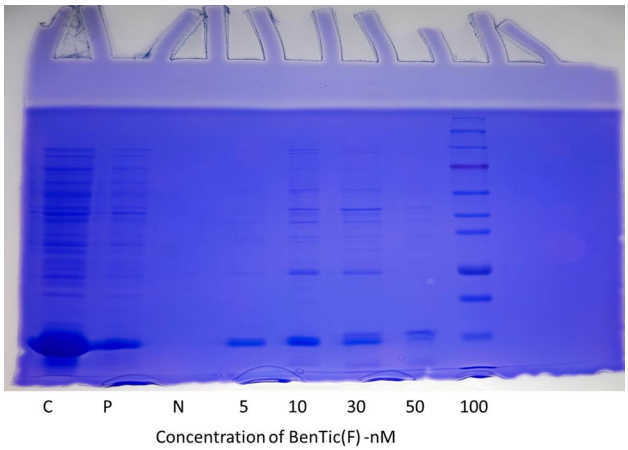


Fig. 8 SDS-PAGE gel showing dose-dependent reduction in phosphorylated MBP bands with increasing BenTic(F) concentrations

through the nonpolar residue Ile220 and the basic residue His221. The MD results clearly indicate that both BenTic(F) and Pazopanib adopt different conformations to maintain their stability in the binding cavity of the protein.

DFT Calculations

Density functional theory (DFT) calculations were performed using the Gaussian 16 software suite to better understand the electronic transitions between the BenTic(F) and the model protein. In the gas phase, monomer and complex were optimised at B3LYP/6–31 g(d, p) level of theory in the gas phase. The optimised structure with specific amino acids in the binding pocket of a given protein and studied using cutting-edge density functional theory.

The Fig. 12 (B) depicts the interaction between BenTic(F) and amino acids. Protein-ligand complexes show O-H···N and N-H···O type hydrogen bonding. Asp (241) demonstrates O-H----N (3.353 Å), Glu (243) demonstrates O-H----N (1.949 Å), which contribute to the complex's stability. The binding energy for BenTic(F) with amino acids was −37.30 kcal/mol, which was higher than the binding energy seen in virtual screening due to the overestimation of the B3LYP method. The variance was attributed to overestimation of binding energy using the B3LYP approach (Ramachandran et al. 2020), and the system was used as the model rather than the entire protein environment. These findings were subsequently studied with frontier molecular orbital (FMO) simulations.

The Fig. 13a and b display the results of frontier molecular orbital computations. The HOMO-LUMO gap in the BenTic(F) (monomer) and BenTic(F)…amino acid complex was found to be 5.13 eV and 4.9 eV, respectively. The reduction in band gap for the complex indicates the stability (Bhaskar et al. 2020).

Global reactivity descriptors such as chemical hardness (η), softness (S), chemical potential (μ), and electrophilicity index (ω) offer critical insights into the electronic stability and reactivity of molecules. These descriptors are derived from the frontier molecular orbital energies, particularly the HOMO-LUMO gap. Chemical hardness quantifies a molecule's resistance to changes in electron density, while softness, defined as the inverse of hardness, reflects its reactivity

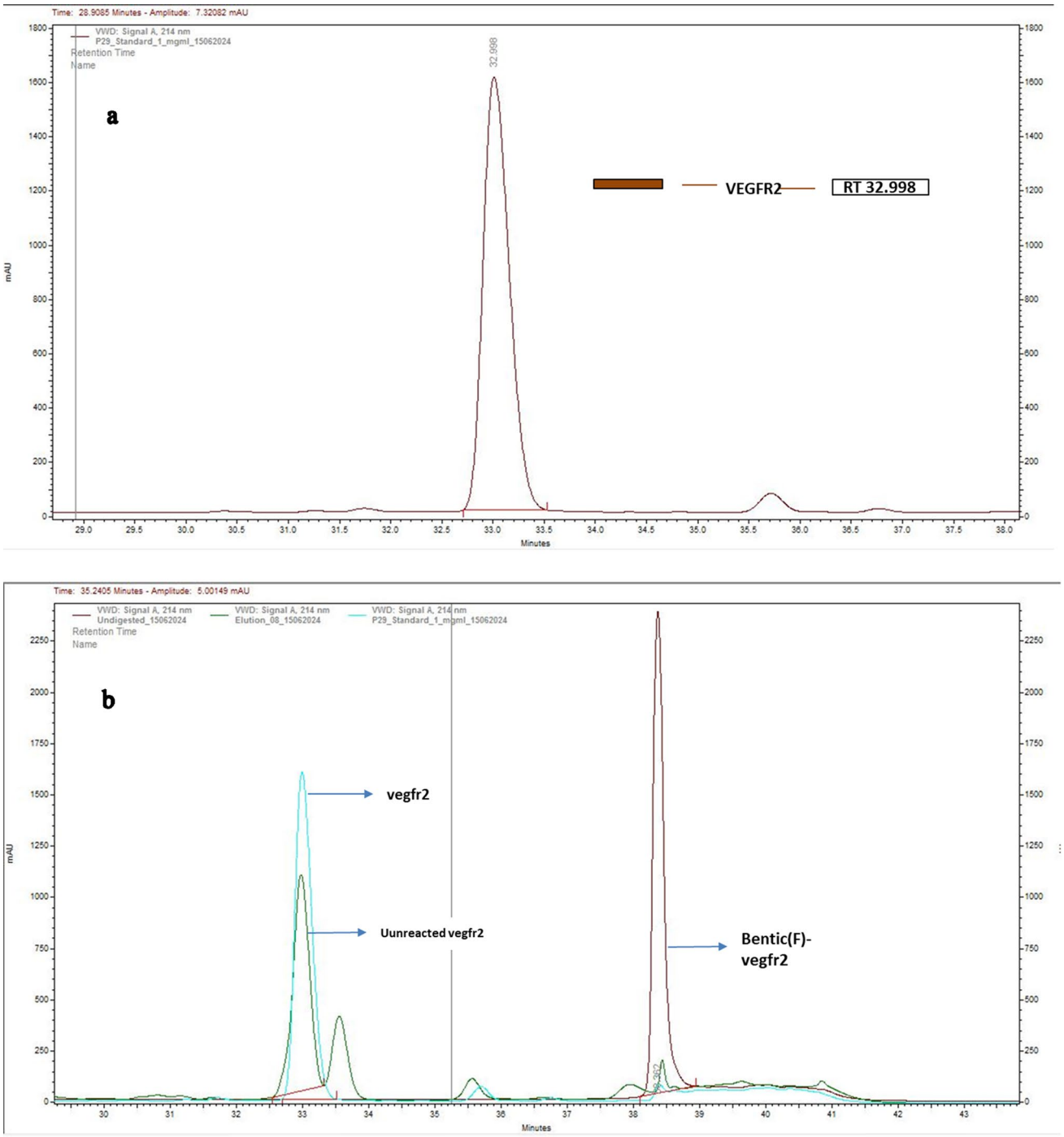


Fig. 9 HPLC validation of VEGFR-2 inhibition. **a** Retention profile of VEGFR-2 under control conditions. **b** Significant reduction in VEGFR-2 peak intensity following treatment with 30 nM BenTic(F), confirming kinase inhibition observed in biochemical assays

and polarizability. The chemical potential indicates the tendency of a molecule to lose electrons and the electrophilicity index represents the potential of the molecule to gain electrons.

As shown in Table 5, the HOMO-LUMO gap for **BenTic(F)** is 5.137 eV, corresponding to a hardness of 0.093 eV and a softness of 10.804 eV^{-1}. The chemical potential and electrophilicity index are calculated as −0.146 eV and 0.115 eV, respectively. Upon interaction with the model 1Y6A protein, a slight change was observed: the HOMO–LUMO gap decreases to 4.937 eV, indicating increased electronic reactivity. The hardness decreases to 0.091 eV, and softness increases to 11.002 eV^{-1}, consistent with enhanced electron delocalisation. As observed in

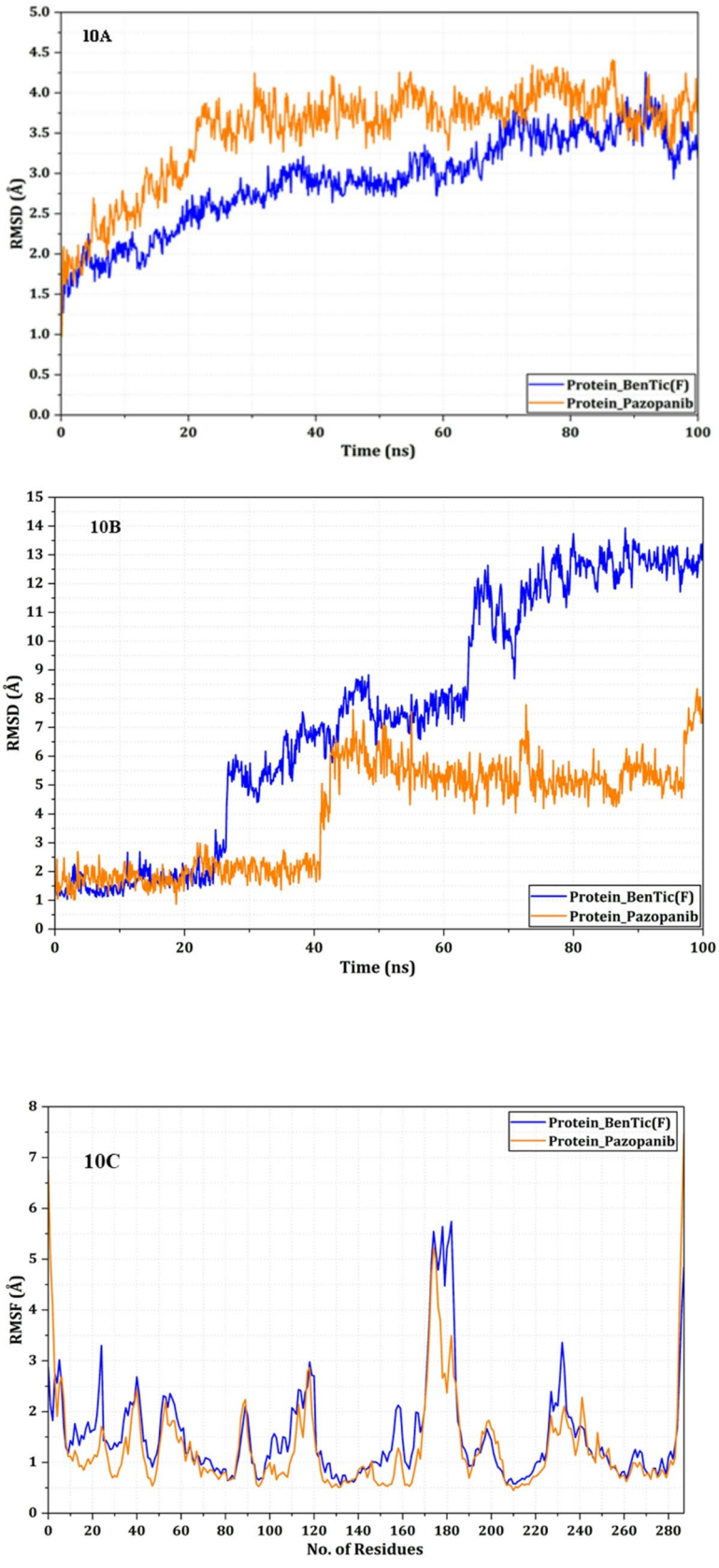


Fig. 10 MD simulation of protein-bound BenTic(F) and Pazopanib complex. The 100 ns MD simulation data were analysed, and the graphs were plotted against time. 10 **A** Protein RMSD, 10 **B** Ligand RMSD, 10 **C** Protein RMSF

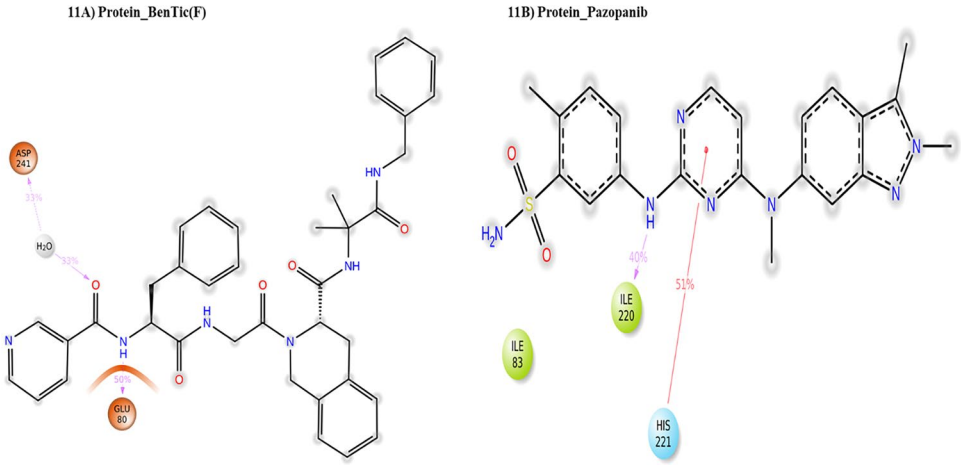


Fig. 11 The percentage of interactions mediated between ligand atoms and protein residues during the 100 ns of MD simulation. The polar amino acid residues that mediate interactions over 30% of the simulation time are shown here

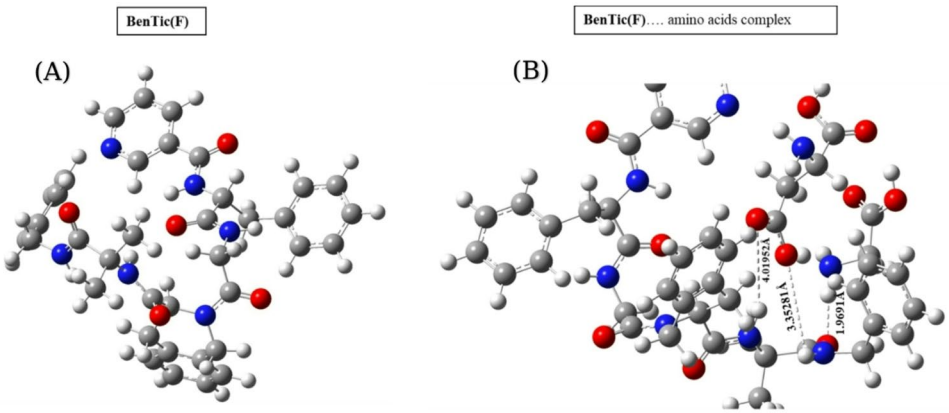


Fig. 12 **A** Optimised structure of BenTic(F), **B** Optimised structure of BenTic(F)…complex with amino acids at the B3LYP/6–31 g(d) level in the gas phase. Distance mentioned in (Å)

Table 5, the chemical potential of BenTic(F)-amino acid complex is slightly less negative than BenTic(F) monomer. The electrophilicity index of the BenTic(F)-amino acid complex has decreased to 0.1096 eV. These shifts suggest that the interaction with amino acid induces moderate changes in the electronic structure of BenTic(F), potentially influencing its reactivity and binding behaviour in a biological environment.

ProTox-3 Predictions

The toxicity predictions indicated significant differences, as shown in Fig. 14 in the safety profiles between BenTic(F) and Pazopanib. There was a low risk of hepatotoxicity and cardiotoxicity for both compounds, but activity in respiratory toxicity raises the possibility of pulmonary side effects. The model predicted neurotoxicity for BenTic(F), but no neurotoxic effects were observed for Pazopanib.

Pazopanib was associated with increased risk for toxicity endpoints, especially carcinogenicity, immunotoxicity, and blood-brain barrier permeability, which may limit its potential for therapeutic use. On the other hand, BenTic(F) had a more encouraging profile in those endpoints, but predicted interactions with PXR and CYP3A4 point toward potential drug-drug interactions. In contrast, BenTic(F) did not cause activation of the p53 stress response pathway; however, Pazopanib did. In terms of metabolism, Pazopanib was predicted to interfere with CYP2C9, while BenTic(F) showed close to critical activation of CYP3A4. Elucidation of nuclear receptor activity shows that both compounds are largely inactive in this pathway, easing doubts about their ability to disrupt the endocrine system.

Such computational findings reveal important variations in toxicity patterns, revealing that Pazopanib carries more safety concerns while emphasising the demand for subsequent experimentation.

DPPH Assay Results

The DPPH radical scavenging assay was used to test whether the synthesised compounds could function as antioxidants, and the results are shown in Fig. 15; Table 6. The strongest antioxidant activity was observed for BenTic(F) as compared with other tested molecules, with an IC_{50} value of 52.19 ± 0.36 µM. This IC_{50} value is observed to be higher than the reference drug pazopanib (88.14 ± 0.57) for all other BenTic analogues. BenTic(S), BenTic(W), BenTic(P),

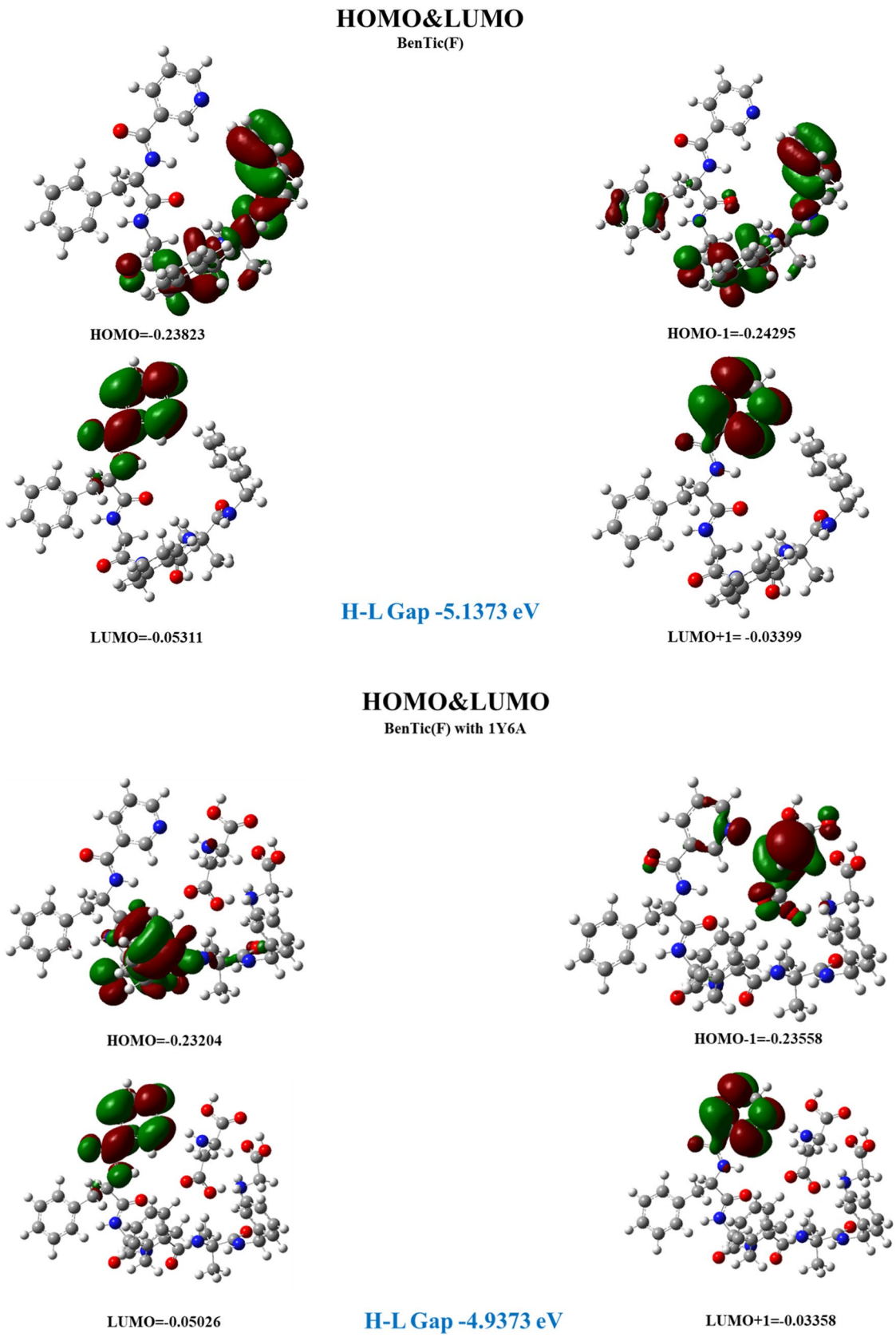


Fig. 13 **a** Electronic transition from the Highest Occupied Molecular Orbital (HOMO) to the Lowest Unoccupied Molecular Orbital (LUMO) in the BenTic(F) **b** Electronic transition from the Highest Occupied Molecular Orbital (HOMO) to the Lowest Unoccupied Molecular Orbital (LUMO) in the complex BenTic(F) amino acid

Table 5 Global reactivity descriptors of BenTic(F) and BenTic(F) with 1Y6A complex

Molecule	H-L Gap (eV)	Hardness (η)(eV)	Softness $1/(\eta)$ $(eV)^{-1}$	Chemical potential (μ) (eV)	Electrophilicity index (ω) (eV)
BenTic(F)	5.137	0.092	10.804	−0.146	0.115
BenTic(F) with amino acid (1Y6A)	4.937	0.090	11.002	−0.142	0.110

and BenTic(H) have much weaker antioxidant effects, with IC_{50} values between 157.26 ± 0.50 μM to 184.60 ± 0.40 μM.

The standard errors of the mean (± 0.36–1.01 μM) for all samples show that the assay results are statistically strong and can be repeated.

Despite ascorbic acid displaying a lower IC_{50} and enhanced DPPH radical scavenging activity, BenTic(F), with an IC_{50} of 34.49 ± 0.24 μg/mL, reveals significant antioxidant potential. BenTic(F) is an intriguing therapeutic candidate due to its dual functionality, demonstrating both effective anti-angiogenic and significant antioxidant capabilities. In contrast to the use of ascorbic acid, BenTic(F) provides a targeted approach towards tumour therapy by synergistically regulating oxidative stress and angiogenesis in the renal cell carcinoma (RCC) tumour microenvironment. Consequently, its antioxidant activity needs to be seen as a complementary asset that magnifies its unique anticancer effects, instead of as a basis for direct comparison with traditional antioxidants like ascorbic acid.

The concentration–response curves in Fig. 15 infer dose-dependent oxidant activity. This suggests BenTic(F) can work as both an anti-angiogenic and an antioxidant agent. This supports its potential for multitargeted therapy in RCC and related diseases.

Conclusion

The 500-novel peptide-(D)-tetrahydroisoquinoline (THIQ) conjugates were designed through structure-based drug design, observed as potential dual-action therapeutics for renal cell cancer (RCC). The current study identified and evaluated five lead candidates with significantly high VEGFR-2 binding affinities (−10.2 to −11.8 kcal/mol) using virtual screening of 500 designed compounds. Furthermore, the highest binding energies of the five compounds observed through HTS were synthesised and analysed using the MTT and DPPH assays, revealing that BenTic(F) is the most effective among the five compounds. BenTic(F) was observed as an improved multi-targeted agent due to its potent VEGFR-2 inhibition (IC_{50} = 47.94 nM) and significant antioxidant activity (DPPH IC_{50} = 52.19 ± 0.36 μM). The molecule exhibits a 2.6-fold superior VEGFR-2 inhibitory efficacy relative to the clinical standard Pazopanib, along with its capacity to mitigate oxidative stress, a critical pathogenic mechanism in RCC.

The strategic peptide conjugation method significantly enhanced the drug solubility in water (HIA = 0.899 compared to 0.015 for pazopanib) while maintaining its drug-like characteristics (LogP = 2.7, TPSA = 139.6 Å²) within Lipinski's parameters. Molecular dynamics simulations lasting 100 ns for BenTic(F) adopt different conformations *via* hydrophilic bonds with critical VEGFR-2 residues (Asp241, Gly243) and strategic hydrophobic interactions (Leu84, Leu214, Phe242) to maintain their stability in the binding cavity of the protein. This indicates that the allosteric site adjacent to hinge engagement and the anchoring of conserved residues may mitigate resistance. An exhaustive ADMET profile indicates that BenTic(F) was safe and reduced toxicity risks for hERG, DILI, and carcinogenicity. The ProTox-3 prediction of potential neurotoxicity requires

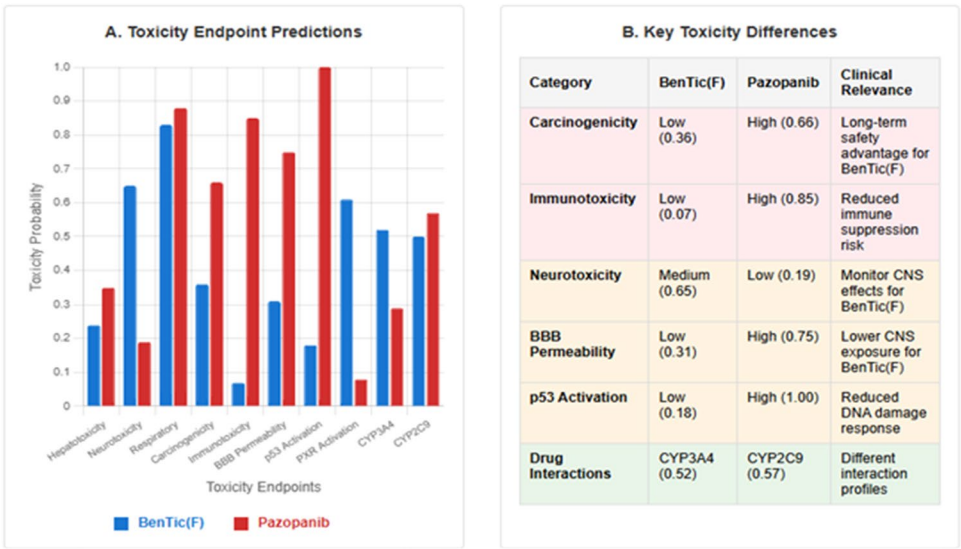


Fig. 14 Comparative in silico toxicity predictions for BenTic(F) and pazopanib across multiple endpoints. **A** Probability scores for key toxicity endpoints showing complementary risk profiles. Values represent predicted probability of toxicity (0–1.0 scale). **B** Summary of clinically relevant differences highlighting BenTic(F)'s improved safety profile in carcinogenicity and immunotoxicity, with manageable trade-offs in neurotoxicity. Colour coding: red (high risk, > 0.6), orange (medium risk, 0.4–0.6), green (low risk, < 0.4). Predictions generated using validated QSAR models. BBB = blood-brain barrier; CYP = cytochrome P450

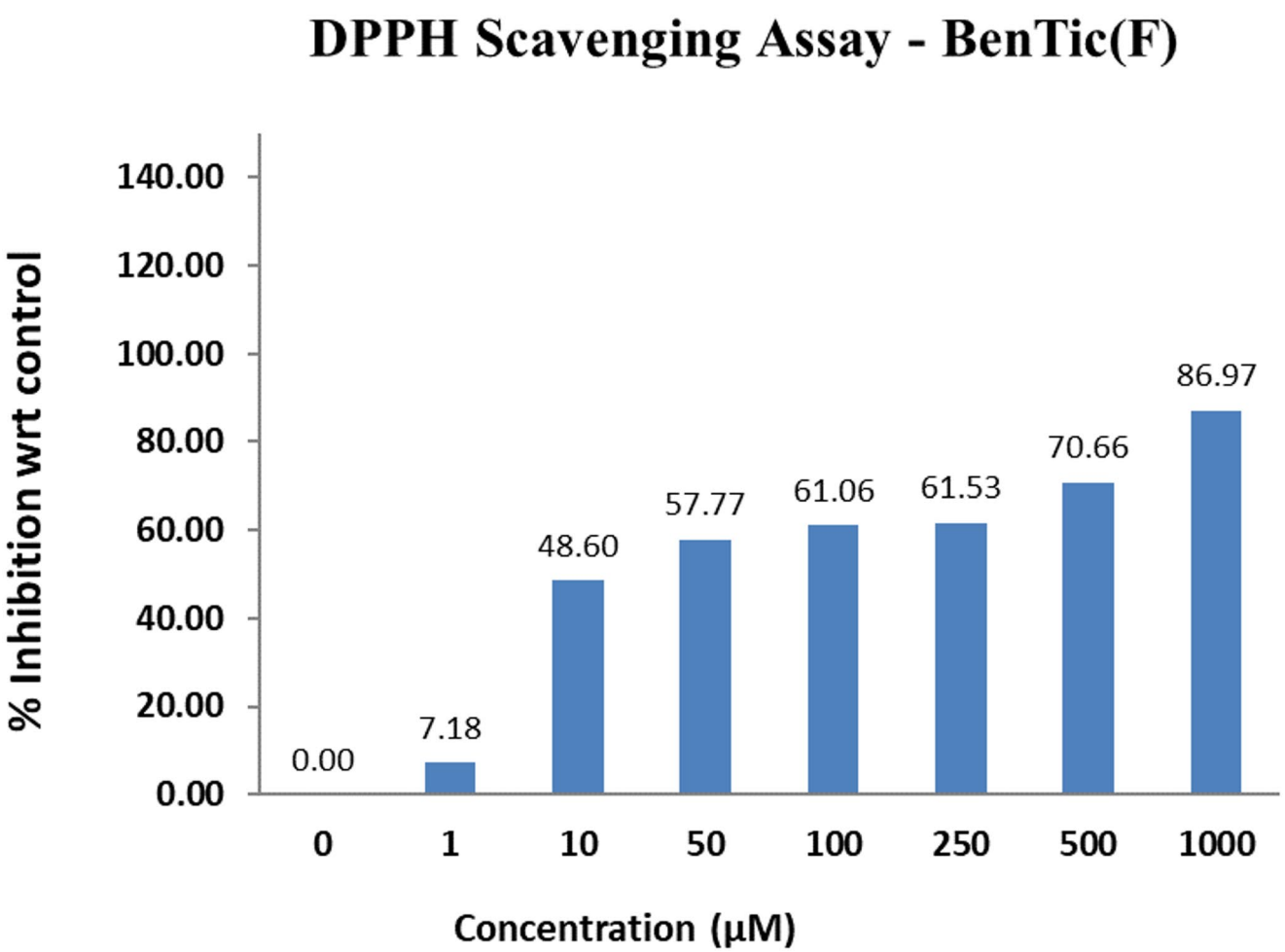


Fig. 15 The calculated IC_{50} values with respect to Control using software Graph Pad Prism − 6

Table 6 IC_{50} values of BenTic derivatives and pazopanib against antioxidant activity

Sample code	IC_{50} value (Mean ± SEM)
BenTic(F)	52.19 ± 0.36 µM
BenTic(S)	172.66 ± 1.01 µM
BenTic(W)	157.26 ± 0.50 µM
BenTic(P)	184.60 ± 0.40 µM
BenTic(H)	170.87 ± 0.36 µM
Pazopanib	88.14 ± 0.57 µM

meticulous examination in forthcoming safety studies. DFT simulations validated the compatibility and stability of hydrophilic Interactions at critical VEGFR-2 model binding sites.

The alignment of computational predictions with virtual molecular docking, kinase assays, cytotoxicity evaluations, and antioxidant assessments provides robust evidence for BenTic(F)'s potential as a pharmaceutical agent. The peptide-THIQ combination represents a significant advancement in the development of precisely designed cancer therapeutics. These pharmaceuticals would focus on both angiogenesis and oxidative stress pathways critical for the progression of renal cell carcinoma (RCC). The current study introduced a promising candidate for RCC therapy and advances in the field of peptide-THIQ conjugates as sophisticated instruments for precision oncology, thereby addressing unmet clinical demands in renal cell carcinoma treatment.

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Data Availability Supporting information.xlsx, Supporting information.Doc, Supporting information.PDF.

Declarations

Competing Interests The authors declare no competing interests.

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